

Timothy Jay M. Marquez

Full Stack Developer / Embedded Systems Engineer

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📍 Puerto Rico St, Makati City, Metro Manila

📅 November 18, 2000

🔗 <https://portfolio-tj.vercel.app/>

Profile

Versatile developer with strong experience across **full-stack web development, embedded systems, automation, and computer vision**. Proven in building scalable web platforms, real-time IoT systems, AI-powered detection workflows, and cloud-connected applications. Skilled in designing maintainable backend services, responsive frontends, and firmware-level hardware control. Comfortable working across the entire technology stack from hardware to cloud.

Technical Skills

Embedded Systems & Firmware Development ● ● ● ● ●

- Raspberry Pi, ESP, Arduino
- C/C++ & Python firmware development
- GPIO, PWM, interrupts, timers
- I2C, SPI, UART communication
- Sensor integration & motor control
- Hardware debugging & reliability testing

Tools & Platforms ● ● ● ● ●

- Linux (Ubuntu, Debian-based systems)
- Windows development environments
- Git & GitHub
- Vercel (frontend deployment & hosting)
- REST API tools (Postman, API testing & debugging)
- Firebase (Auth, Realtime DB, cloud services)
- Supabase (PostgreSQL backend, storage, auth)
- MySQL, PostgreSQL
- WebSockets & real-time messaging
- Bash / terminal workflows
- Logging, debugging & monitoring tools

Full-Stack & Software Development ● ● ● ● ●

- Backend APIs & system architecture
- Frontend component-based development
- RESTful services & real-time systems
- Authentication, role-based access control
- Database design & optimization
- Cloud-connected applications

AI, Data & Computer Vision ● ● ● ● ●

- YOLO (v3–v9) real-time object detection & custom model training
- PyTorch-based deep learning pipelines
- OpenCV for image capture, preprocessing, and video streams
- Dataset collection, labeling, and augmentation
- Model evaluation & performance tuning
- Real-time inference on edge devices (Raspberry Pi, embedded systems)
- NumPy for numerical computing
- Pandas for data cleaning, transformation, and analysis
- Image thresholding, contour detection, and feature extraction
- Automation triggers based on AI detection results

Professional Experience

2022 – Present
Tuguegarao City,
Cagayan, Philippines

Embedded & Full-Stack Systems Developer

Upwork / Independent / Academic Projects

- Designed, developed, and deployed embedded and full-stack systems integrating hardware, firmware, backend services, and web interfaces
- Developed firmware for microcontrollers and single-board computers using **C/C++ and Python**, including GPIO control, sensor interfacing, motor control, and real-time logic
- Integrated cameras, sensors, actuators, and automation workflows with **RESTful APIs** and web-based dashboards

- Built responsive frontend interfaces for device monitoring and control using modern web technologies
- Designed and implemented scalable backend services with **authentication, role-based access, data modeling, logging, and error handling**
- Implemented cloud-connected systems for **real-time data acquisition, processing, and visualization**
- Worked with databases for structured data storage, querying, and performance optimization
- Applied **debugging, profiling, testing, and optimization** techniques across hardware, firmware, and software layers
- Deployed and maintained applications on **Linux-based environments**, including server configuration and process management
- Integrated third-party APIs and services for notifications, data exchange, and system automation
- Followed best practices in **version control (Git/GitHub)**, documentation, and iterative development

10/2024 – 03/2025
Tuguegarao City,
Cagayan

Laboratory Technician & Part-Time Instructor

Cagayan State University Carig

- Assisted in laboratory classes involving electronics, embedded systems, and computer engineering fundamentals
- Prepared, tested, and maintained laboratory equipment, tools, and instructional materials
- Guided students in hands-on experiments, circuit assembly, and troubleshooting
- Provided technical support during laboratory sessions and practical examinations
- Assisted in instruction, demonstrations, and supervision for engineering-related courses
- Ensured laboratory safety, proper equipment usage, and operational readiness

10/2023 – 03/2024
Tuguegarao City,
Cagayan

Head Technician

Silicon Valley – Tuguegarao

- Performed hands-on diagnosis, repair, and maintenance of computer systems and consumer electronics
- Troubleshoot hardware and software issues on desktops, laptops, and peripherals
- Installed, configured, and upgraded operating systems, drivers, and applications
- Conducted hardware replacements, system testing, and quality checks
- Documented repair work and ensured devices were fully functional before release
- Provided technical explanations and support to customers regarding repairs

Education

08/2019 – 07/2023
Tuguegarao City,
Cagayan

Batchelor of Science in Computer Engineering

Cagayan State University Carig